

STUDENT PERFORMANCE PREDICTION SYSTEM

End-to-End Machine Learning Project Report



Submitted as part of
AI/ML Internship



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1. Introduction

Machine Learning is a branch of Artificial Intelligence that enables computers to learn patterns from data and make predictions without being explicitly programmed for every task.

Student academic performance can be influenced by several factors such as study hours, attendance and performance in other subjects. In this project, Machine Learning techniques are used to analyze student performance data and predict Mathematics scores.

The project follows an end-to-end Machine Learning workflow including data collection, data cleaning, exploratory data analysis, preprocessing, feature selection, model training, model evaluation and prediction.

2. Problem Statement

Educational institutions collect different types of student performance information. However, manually analyzing this information to understand and predict academic performance can be difficult.

The problem addressed in this project is to develop a Machine Learning based system that predicts a student's Mathematics score using available demographic and academic features such as age, study hours, attendance, Science score, English score and gender.

3. Objectives

- Collect and understand student performance data.
- Clean the dataset and handle missing values.
- Perform Exploratory Data Analysis (EDA).
- Identify important features for prediction.
- Preprocess the data for Machine Learning.
- Train regression models.
- Evaluate and compare model performance.
- Develop a simple student score prediction system.
- Analyze the results and identify future improvements.

4. Technologies Used

Technology	Purpose
Python	Programming language
Pandas	Data manipulation
NumPy	Numerical operations
Matplotlib	Data visualization
Scikit-learn	Machine Learning
Joblib	Model saving
Jupyter Notebook	Development environment

5. Dataset Description

The project uses a Student Performance dataset containing demographic and academic information about students.

The main attributes include Student ID, Name, Gender, Age, Study Hours, Attendance, Mathematics Score, Science Score and English Score.

The Mathematics Score is selected as the target variable for the prediction task.

Main Features:

- Age
- Study Hours
- Attendance
- Science Score
- English Score
- Gender

Target Variable:

- Mathematics Score

6. Data Preprocessing

Data preprocessing was performed before training the Machine Learning models.

The following preprocessing steps were applied:

1. Missing values were identified and handled using median values for numerical columns.
2. Duplicate records were identified and removed.
3. The categorical Gender feature was converted into numerical form using Label Encoding.
4. The dataset was divided into training and testing sets.
5. StandardScaler was used to scale the features for Linear Regression.

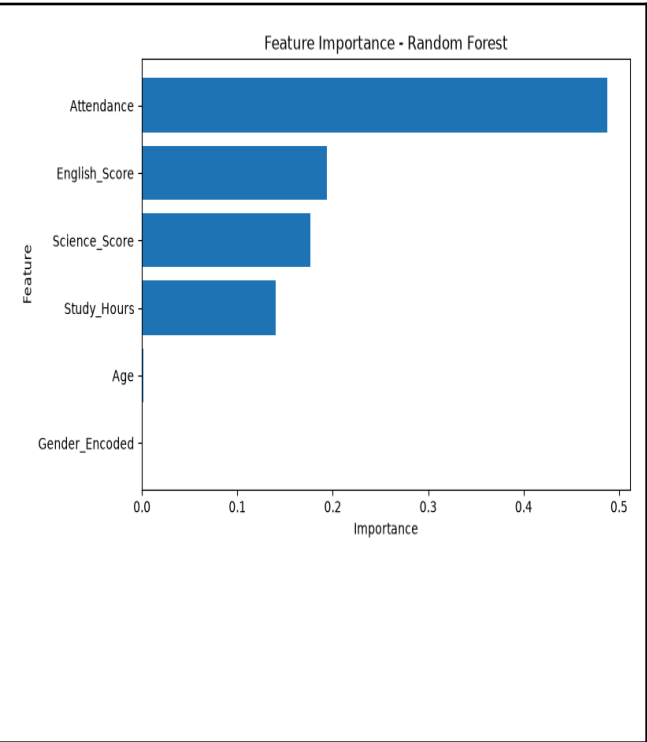
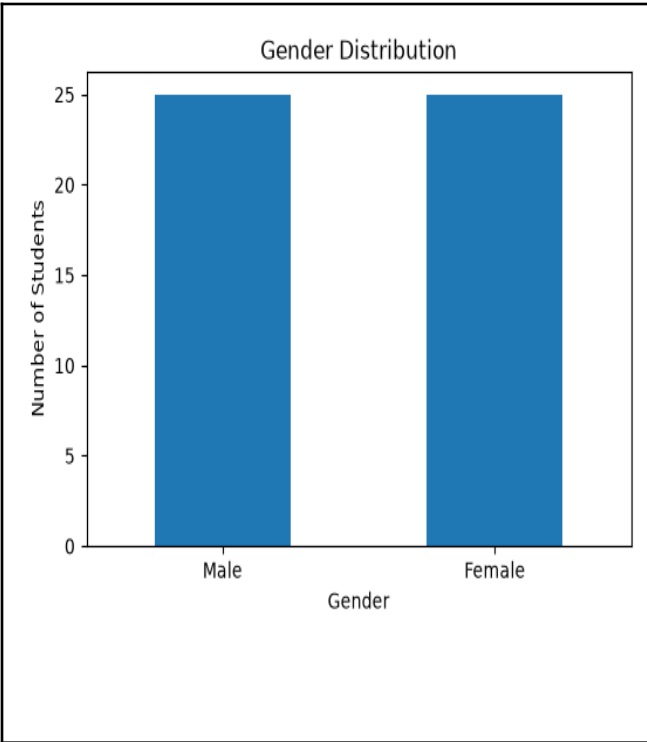
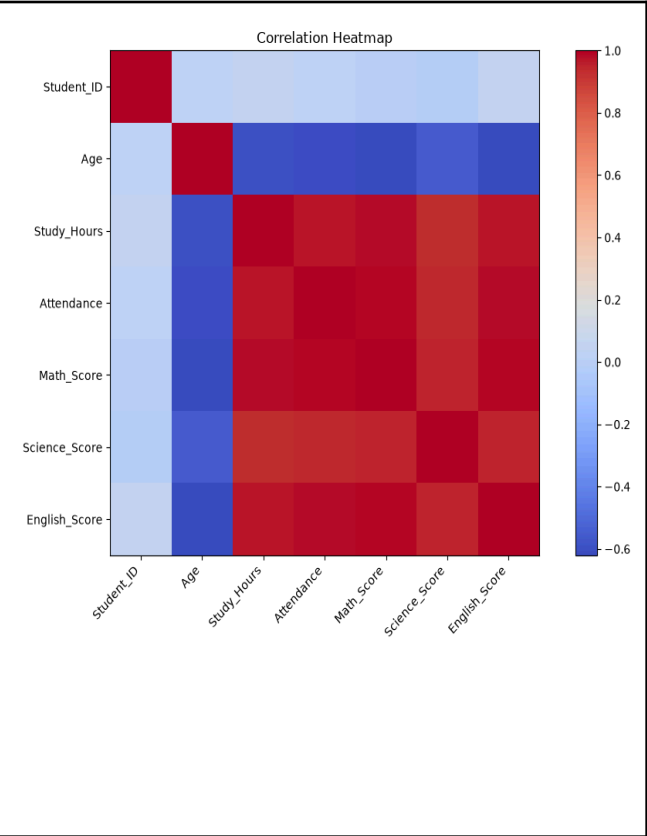
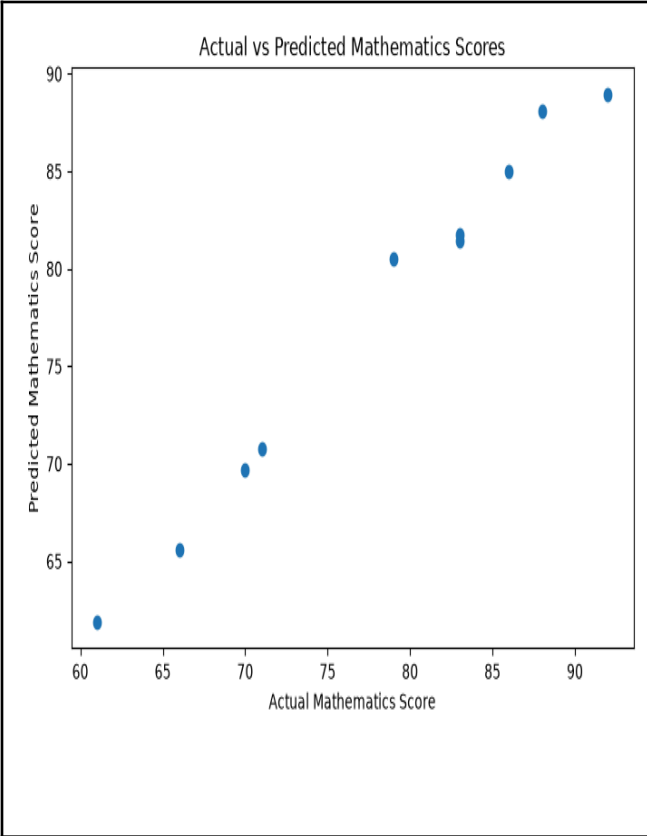
7. Exploratory Data Analysis

Exploratory Data Analysis was performed to understand patterns and relationships in the dataset.

The analysis included:

- Gender distribution
- Study hours distribution
- Attendance distribution
- Subject score distributions
- Study Hours vs Mathematics Score
- Correlation analysis
- Feature importance analysis

Visualizations were created using Matplotlib to understand the data more effectively.



8. Model Development

Two supervised Machine Learning regression algorithms were developed for predicting Mathematics scores.

1. Linear Regression

Linear Regression was used as a baseline regression model for predicting the Mathematics score based on the selected features.

2. Random Forest Regression

Random Forest Regression was used as a second model. It combines multiple decision trees and can capture non-linear relationships between input features and the target variable.

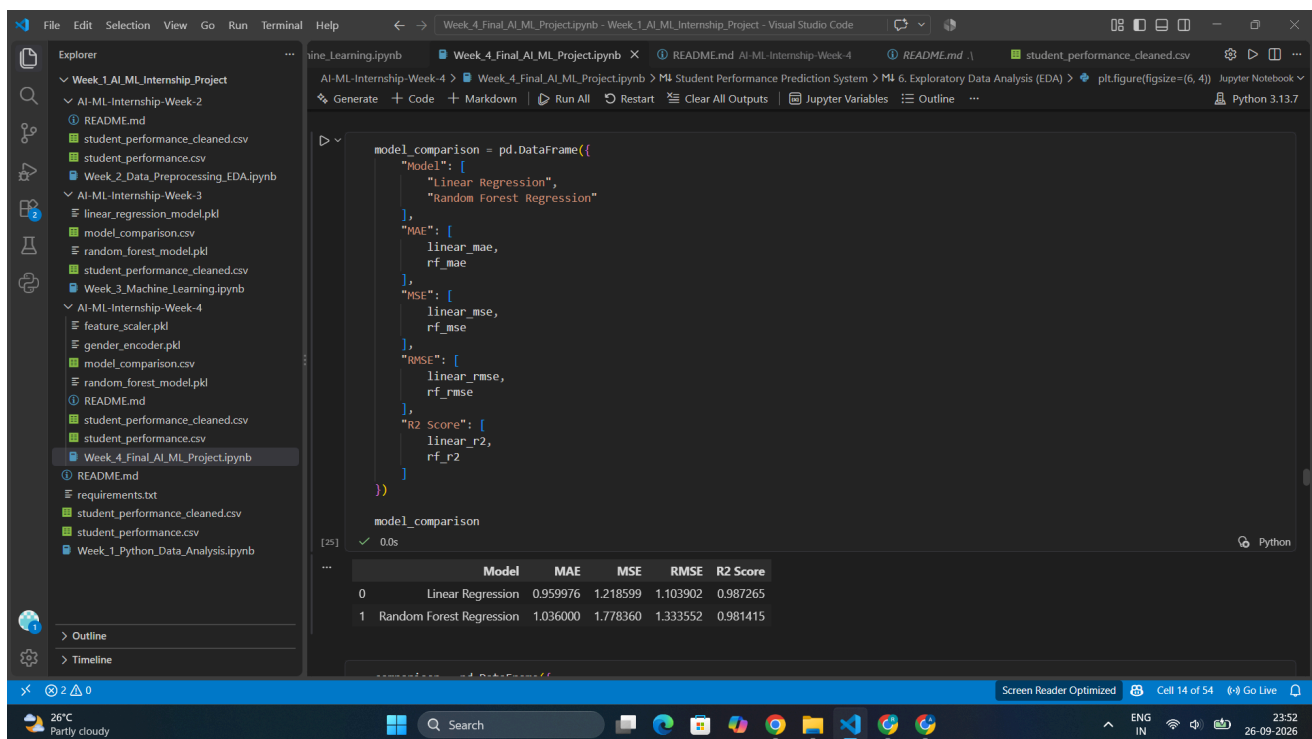
Both models were trained using the training dataset and tested on unseen test data.

9. Model Evaluation

The trained models were evaluated using four standard regression metrics:

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- R^2 Score

MAE, MSE and RMSE measure prediction error, while R^2 Score measures how well the model explains variation in the target variable.



The screenshot displays a Jupyter Notebook environment within Visual Studio Code. The notebook is titled 'Week_4_Final_AI_ML_Project.ipynb' and is part of a project named 'AI-ML-Internship-Week-4'. The left sidebar shows the Explorer view with various files and folders, including 'student_performance_cleaned.csv', 'Week_2_Data_Preprocessing_EDA.ipynb', and 'Week_3_Machine_Learning.ipynb'. The main editor area shows a Python code cell with the following code:

```
model_comparison = pd.DataFrame({
    "Model": [
        "Linear Regression",
        "Random Forest Regression"
    ],
    "MAE": [
        linear_mae,
        rf_mae
    ],
    "MSE": [
        linear_mse,
        rf_mse
    ],
    "RMSE": [
        linear_rmse,
        rf_rmse
    ],
    "R2 Score": [
        linear_r2,
        rf_r2
    ]
})

model_comparison
```

The output of the code cell is a DataFrame with the following data:

	Model	MAE	MSE	RMSE	R2 Score
0	Linear Regression	0.959976	1.218599	1.103902	0.987265
1	Random Forest Regression	1.036000	1.778360	1.333552	0.981415

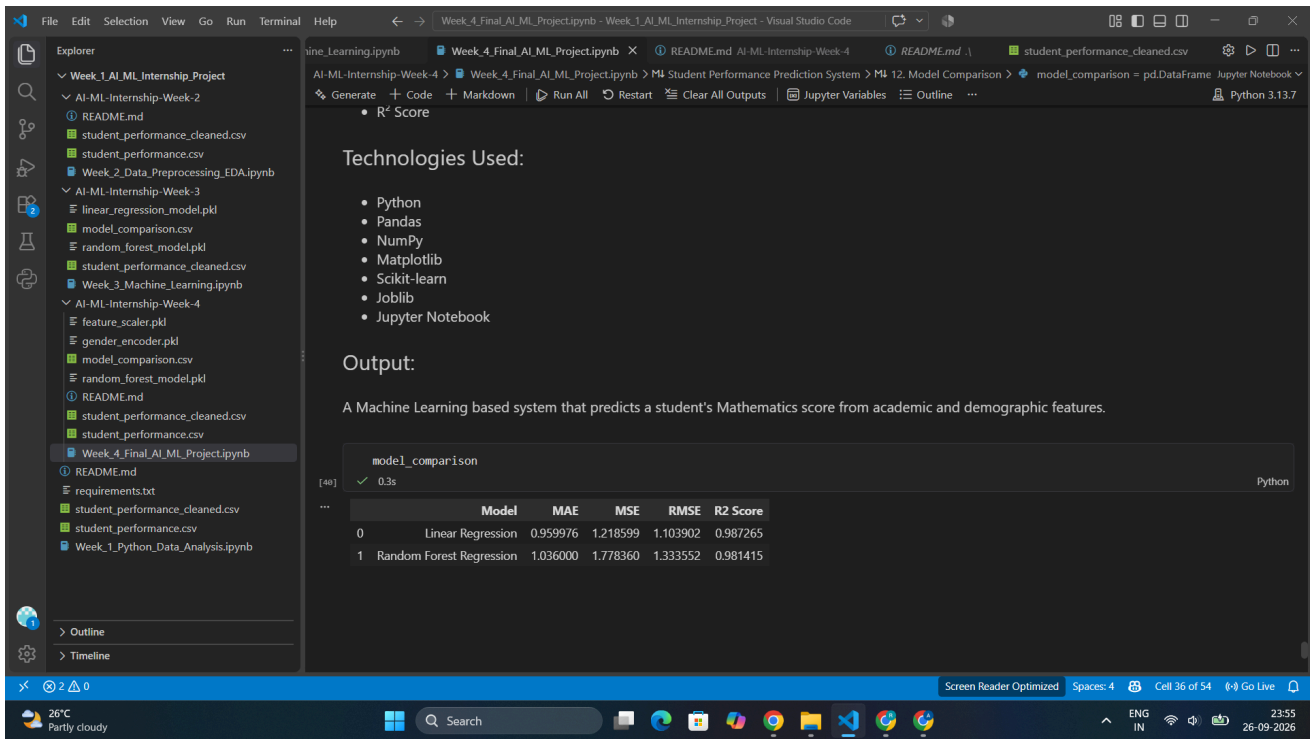
The bottom status bar of the application shows 'Screen Reader Optimized', 'Cell 14 of 54', and 'Go Live' button. The system tray at the bottom indicates '26°C', 'Partly cloudy', and the date '26-09-2026'.

10. Results

The project successfully trained two Machine Learning regression models for Mathematics score prediction.

The performance of Linear Regression and Random Forest Regression was compared using MAE, MSE, RMSE and R^2 Score.

The model comparison results were saved in a separate CSV file. The final prediction system uses the trained Random Forest model to generate Mathematics score predictions for new student information.



The screenshot displays a Jupyter Notebook titled "Week_4_Final_AI_ML_Project.ipynb" within the Visual Studio Code interface. The notebook is open to a cell containing a table of model comparison results. The table compares Linear Regression and Random Forest Regression models based on MAE, MSE, RMSE, and R^2 Score. The Random Forest model shows superior performance with lower error metrics and a higher R^2 score.

Technologies Used:

- Python
- Pandas
- NumPy
- Matplotlib
- Scikit-learn
- Joblib
- Jupyter Notebook

Output:

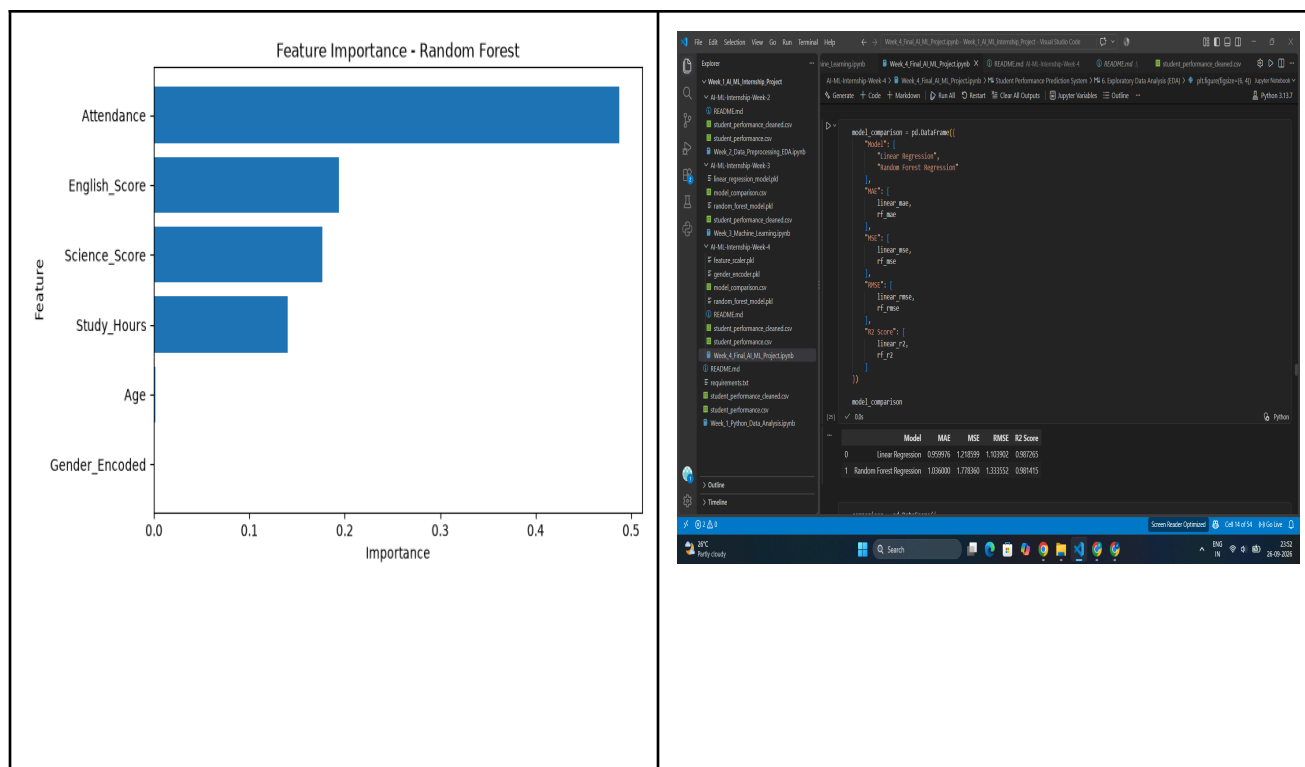
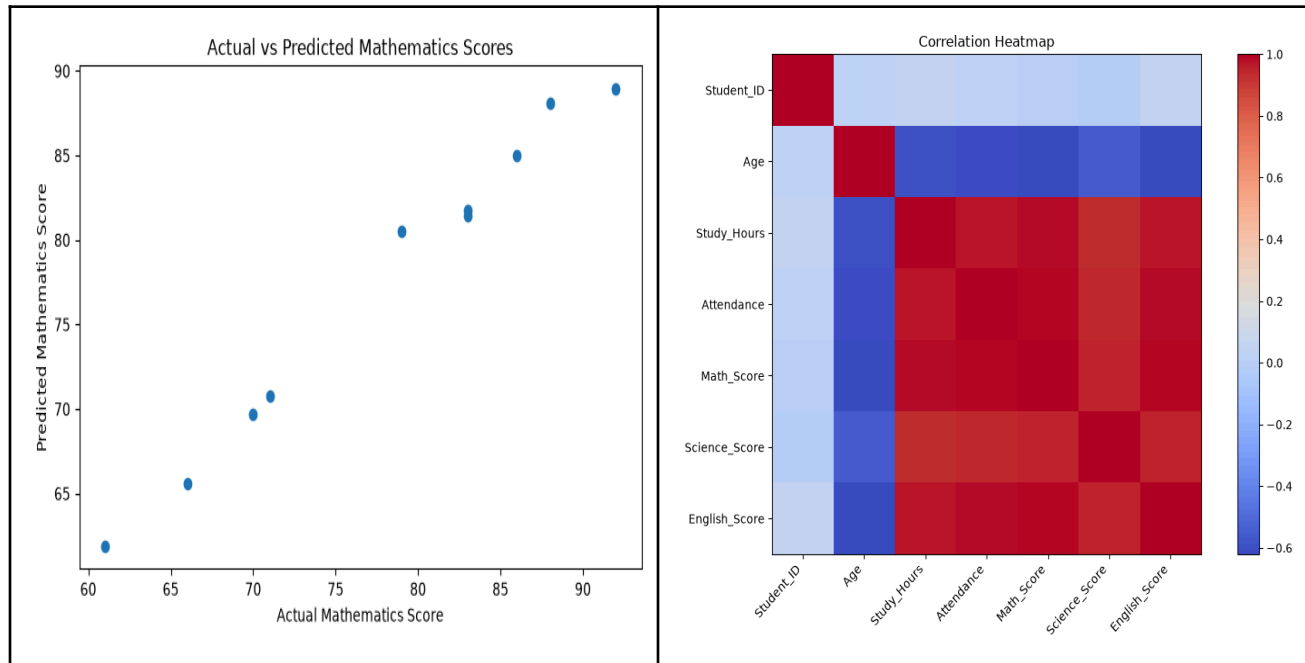
A Machine Learning based system that predicts a student's Mathematics score from academic and demographic features.

```
model_comparison
```

	Model	MAE	MSE	RMSE	R^2 Score
0	Linear Regression	0.959976	1.218599	1.103902	0.987265
1	Random Forest Regression	1.036000	1.778360	1.333552	0.981415

The bottom status bar of the application shows the system temperature as 26°C, the date as 26-09-2026, and the time as 23:55.

11.Screenshots



12. Challenges Faced

During the development of the project, several challenges were encountered.

- Handling missing values in the dataset.
- Removing duplicate records.
- Converting categorical data into numerical form.
- Selecting suitable features for the prediction task.
- Understanding and comparing different regression models.
- Implementing the prediction system using the trained model.

These challenges helped improve the understanding of the complete Machine Learning workflow.

13. Future Improvements

The project can be further improved in several ways:

- Use a larger and more diverse dataset.
- Include additional academic and behavioral features.
- Test additional Machine Learning algorithms.
- Perform hyperparameter tuning.
- Develop a web interface using Streamlit or Flask.
- Deploy the application on a cloud platform.
- Continuously update the model using new student data.

14. Conclusion

This project successfully developed an end-to-end Machine Learning system for predicting student Mathematics scores.

The project covered the complete Machine Learning workflow, including data collection, data cleaning, exploratory data analysis, preprocessing, feature selection, model development, model evaluation and prediction.

Linear Regression and Random Forest Regression were implemented and evaluated using standard regression metrics.

The project demonstrates how Machine Learning can be applied to student performance data to build a practical prediction system.